

## TITLE

Overall and Heterogeneous Effects of Digital Health Messages on Sleep, Activity, and Mood in Medical Interns

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## INTRODUCTION AND AIMS OF THE STUDY

This section provides an overview of the context of the research question, including what is known and unknown, and requires a detailed literature search. This should contain the following four elements, and will be used to construct the Introduction and Discussion sections of the manuscript:

1. Define the problem or research question.
2. Describe what is known about the research relative to what you plan to study. For example, what studies have already examined an aspect of this research question, and what remaining questions exist that this research project will address.
3. What are the gaps in knowledge that this study will address (usually 2–3).
4. Describe the hypotheses of the study questions.

Although mobile health push notifications aim to improve mood, sleep, and activity, individual responsiveness can vary substantially. NeCamp et al. [1] evaluated notification effects in the 2018 Intern Health Study using Weighted and Centered Least-Squares (WCLS), but WCLS averages effects across participants and decision points, causing opposing individual responses to cancel out and masking true heterogeneity [2]. Furthermore, the trial design has since expanded to deploy diverse therapeutic notification strategies whose individual-level efficacy remains unexamined. Leveraging DR-WCLS pseudo-outcomes [3], this study addresses these gaps by evaluating whether distinct therapeutic strategies exert causal excursion effects on proximal outcomes, testing whether individual treatment responsiveness exhibits significant heterogeneity across participants, and identifying baseline

characteristics that predict positive versus negative intervention responsiveness.

#### DELIVERABLE TYPE AND DEADLINE (OPTIONAL)

|       |                               |                             |
|-------|-------------------------------|-----------------------------|
| _____ | Abstract                      | Deadline Date: _____        |
| _____ | Manuscript                    | Deadline Date: _____        |
| X     | Internal Report               | Deadline Date: Oct 21, 2026 |
| _____ | Pilot data for grant          | Deadline Date: _____        |
| _____ | Other (please explain): _____ |                             |
|       | Deadline Date: _____          |                             |

#### STUDY COHORT

##### PROMPT and COMPASS Cohorts Diagram

|                                                     |                                                     |
|-----------------------------------------------------|-----------------------------------------------------|
| <input type="checkbox"/> RCT PROMPT                 | <input type="checkbox"/> Ro1 PROMPT                 |
| <input type="checkbox"/> COMPASS                    | <input type="checkbox"/> Full Cohort                |
| <input checked="" type="checkbox"/> 2025 IHS Cohort | <input checked="" type="checkbox"/> 2024 IHS Cohort |

We will prioritize 2025 IHS cohort, possibly extending to the 2024 cohort if time permits.

#### DATA TYPES (E.G. SURVEY, FITBIT, GENOMIC, SENSORKIT)

Survey (for BL characteristics), Fitbit & Garmin & HealthKit (for steps/sleep/mood outcomes as well as moderators)

#### INCLUSIONS (E.G. SCREENING CUTOFFS, APP ASSIGNMENT)

N/A

## EXCLUSIONS (IF ANY)

To ensure reliable estimation of participant-level treatment effects, participants with excessive missingness with respect to proximal outcome or moderators will be excluded prior to analysis based on a predefined completion threshold. For retained participants, sporadic decision-point level missingness will be addressed through the observation weighting in the DR-WCLS estimation framework.

## VARIABLES

### MAIN EXPOSURE

Notification (sending notification vs not sending notification), notification type (sleep vs mood vs activity), notification strategy (therapeutic strategies)

### PRIMARY OUTCOME

Self-reported mood, sleep duration, and step count

### SECONDARY OUTCOME (S)

N/A

### ADDITIONAL COVARIATES (PLEASE SPECIFY IF THESE WILL BE CONSIDERED AS CONFOUNDERS, OR MEDIATING AND MODERATING RELATIONSHIPS WILL ALSO BE EXPLORED)

Additional covariates consist of dynamic, time-varying contextual variables measured at each decision point (e.g., previous night's sleep duration). These variables will be evaluated as candidate moderators to assess contextual heterogeneity in treatment responsiveness and determine how real-time user state affects the intervention effect on the proximal outcome.

## STATISTICAL ANALYSIS

*This can be your fully established analysis, or draft prior to potential consult with our analysis team.*

Univariate approach:

Multivariable approach:

We evaluate overall treatment efficacy and individual response heterogeneity using a multi-stage causal inference framework. We begin by applying Doubly Robust Weighted and Centered Least Squares to estimate the overall marginal causal excursion effect across all participants and decision points. Next, using the per-decision-point doubly robust pseudo-outcomes generated by this procedure, we fit participant-specific regressions to estimate individual baseline treatment effects. To establish whether these individual treatment effects are statistically significant overall across the population, we compare observed effect magnitudes against a randomization null distribution generated by permuting treatment assignments under the design probabilities. We then inspect the distribution of estimated participant effects and fit Gaussian Mixture Models to evaluate response heterogeneity and identify distinct responder subgroups. Finally, if significant heterogeneity is confirmed, we regress participant baseline characteristics against individual treatment effects to determine which baseline profiles predict positive versus negative intervention responsiveness.

All analyses will be conducted across the complete study sample as well as stratified by participant device type to evaluate potential hardware-level variability in treatment estimates.

#### LIMITATIONS OF THE STUDY (OPTIONAL)

A primary limitation of this study concerns the preprocessing and harmonization of intensive time-series data across heterogeneous commercial devices. Because participants were not explicitly differentiated by device model, establishing uniform measurement baselines across hardware platforms proved challenging. Consequently, cross-device data harmonization relied on ad-hoc alignment procedures, which may introduce calibration noise or unmeasured sensor variance. On the other hand, device stratification will reduce the power of the analysis. Furthermore, the sensitivity of causal estimates to specific time-series preprocessing choices remains a potential source of analytical variability.

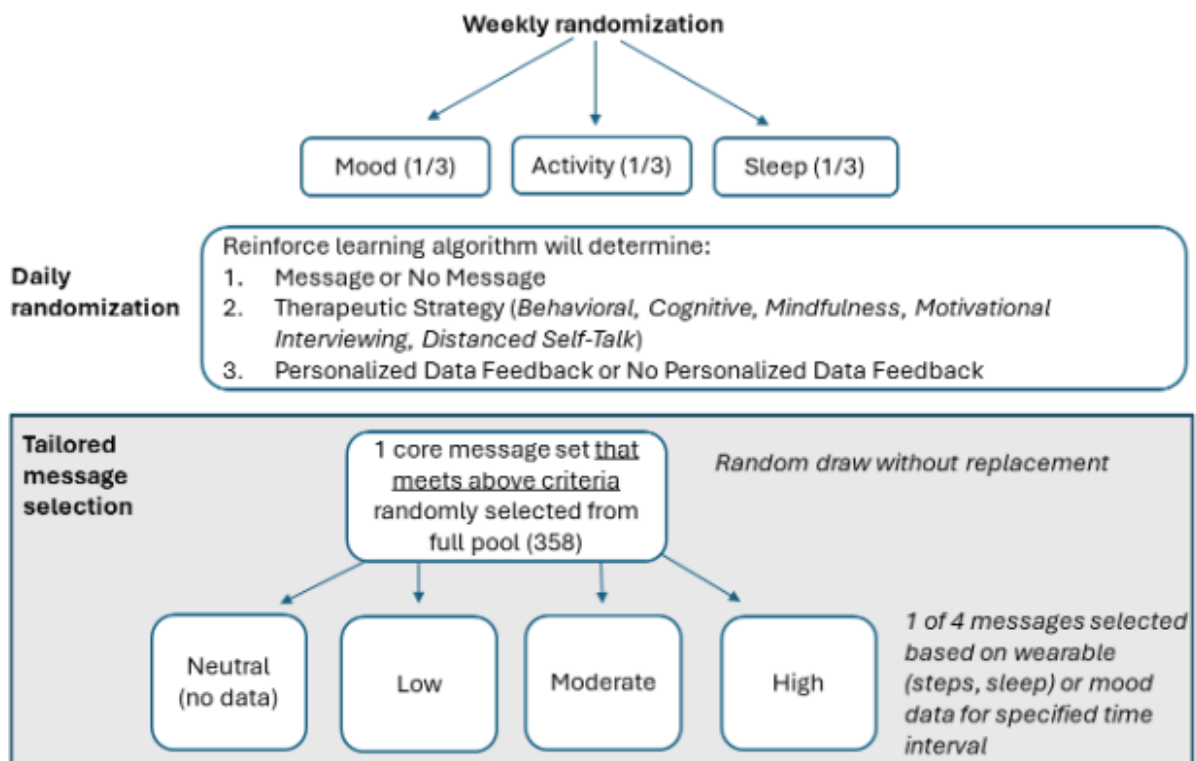
#### TABLES/ FIGURES

Title, description and shell of tables and mock figures. The following are examples for Table 1 and figures.

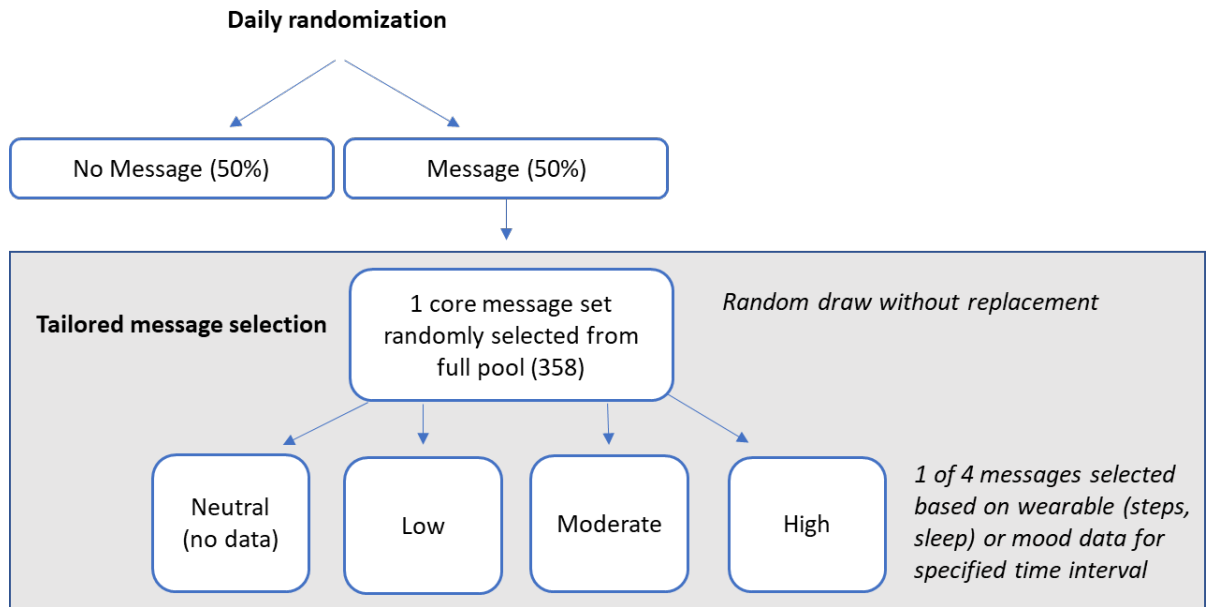
1. Table 1 – Characteristics

| Characteristic | Overall (No. of Patients, %) | Garmin (No. Patients, % Total) | HealthKit (No. Patients, % Total) | Fitbit (No. Patients, % Total) |
|----------------|------------------------------|--------------------------------|-----------------------------------|--------------------------------|
| Race           |                              |                                |                                   |                                |
| White          |                              |                                |                                   |                                |
| Black          |                              |                                |                                   |                                |
| Others         |                              |                                |                                   |                                |
| Age            |                              |                                |                                   |                                |
| ...            | ...                          | ...                            | ...                               | ...                            |

2. Figure 1 – 2025 IHS Study Design



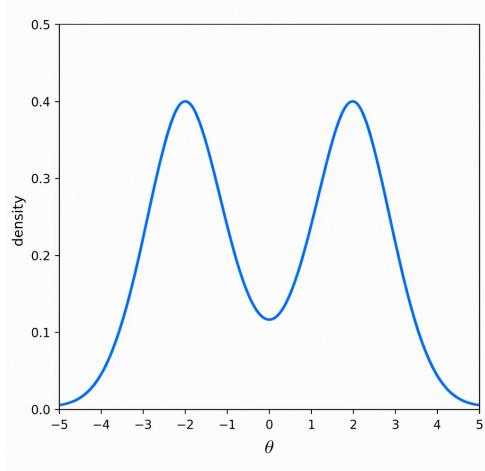
3. Figure 2 – 2024 IHS Study Design



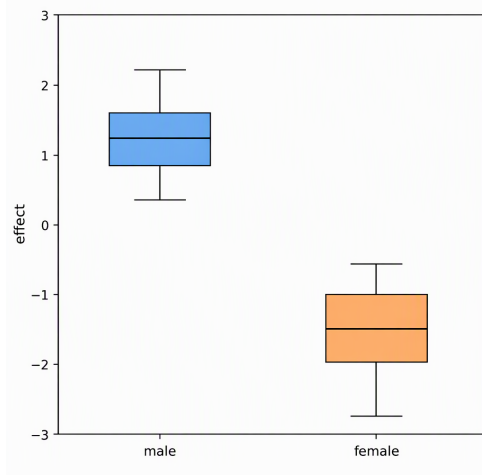
4. Table 2 – Table of Cumulative Magnitude of Effects and P-Values

| Intervention    | Cumulative Magnitude of Effect | P-Value |
|-----------------|--------------------------------|---------|
| Notification    |                                |         |
| Mindfulness     |                                |         |
| ...             | ...                            | ...     |
| No Notification |                                |         |

5. Figure 3 – Distribution of Effects



6. Figure 4 – Effect vs Baseline Characteristic



## REFERENCES

- [1] Timothy NeCamp, Srijan Sen, Elena Frank, Maureen A Walton, Edward L Ionides, Yu Fang, Ambuj Tewari, and Zhenke Wu. Assessing real-time moderation for developing adaptive mobile health interventions for medical interns: Micro-randomized trial. *Journal of Medical Internet Research*, 22(3):e15033, March 2020.
- [2] Tianchen Qian, Ashley E. Walton, Linda M. Collins, Predrag Klasnja, Stephanie T. Lanza, Inbal Nahum-Shani, Mashfiqui Rabbi, Michael A. Russell, Maureen A. Walton, Hyesun Yoo, and Susan A. Murphy. The microrandomized trial for developing digital interventions: Experimental design and data analysis considerations. *Psychological Methods*, 27(5):874–894, October 2022.
- [3] Jieru Shi and Walter Dempsey. A meta-learning method for estimation of causal excursion effects to assess time-varying moderation. *Biometrics*, 81(4), October 2025.